Praneeth Sai Chunduru

Full-stack Python developer

About 4 years in data engineering, AI/RAG systems and multi-tenant SaaS.

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SUMMARY

Results-driven Python developer with 4 years of experience across data engineering, ETL pipelines, system integration and multi-tenant SaaS platforms. I build secure, scalable backends with FastAPI and PostgreSQL, with hands-on expertise in retrieval-augmented generation (RAG), vector databases, LLM integration and AI/ML models for anomaly detection and intelligent data processing.

At SRS, I build features on Luceford, a fintech debt-settlement and collections platform, and oversee development of the SRS Conversational AI Assistant, a multi-tenant RAG chatbot platform, where I focus on reviewing and verifying the backend. The work spans React/TypeScript frontends, PII encryption, RBAC, microservice architecture and CI/CD on AWS.

Core competencies: RAG and vector search · LLM integration and prompt engineering · FastAPI and REST API design · Multi-tenant SaaS architecture · RBAC and application security · PostgreSQL and pgvector · React and TypeScript · Data engineering and ETL · CI/CD and AWS · Automated testing (pytest)

EXPERIENCE

SRS Business Solutions

June 2026 – present

Luceford

Multi-tenant fintech debt-settlement and collections platform

- Core contributor on a platform managing the full client lifecycle: lead intake, creditor negotiation and settlement, ACH collection from client escrow accounts and RPPS disbursement to creditors, under SOX/ICFR-grade financial controls, audit logging and multi-tenancy.
- One of the most active developers on a ~10-person engineering team throughout the build, working across a modular FastAPI backend and several React frontends that share a common component library.
- Built the self-service lead portal end to end, including its approval gate and reporting.
- Built client portal messaging with real email delivery, automated requests, search and attachments.
- Built the Compliance Officer Portal: staff training tracking, licensure management, document review (shared reusable upload component, dynamic category and proof-type rendering, per-document approval), Lead ID search, review-queue sorting and role invitations. Resolved segregation-of-duties access-control violations for users holding overlapping Counselor and Compliance Officer roles.
- Implemented field-level encryption for SSNs and other PII, and staff Employee ID handling, to protect sensitive client and staff data.
- Led a microservice extraction-and-reintegration effort: split the client-portal backend into an independent service with bidirectional sync, then merged it back into the monolith along with the corresponding frontend split.
- Fixed account-collision bugs between staff and client logins in the authentication system.
- Resolved Alembic migration-chain conflicts in a large, fast-moving schema, and worked on cron-based payment scheduling with state-specific fee rules.
- Configured PostgreSQL connectivity across multiple AWS EC2 instances and proposed a migration path to Amazon RDS for long-term scalability.
- Researched the debt-relief domain, including MMI's DMP enrollment flow, the ACH/disbursement lifecycle (NACHA, RPPS, FairShare, BillerID) and US banking UX standards (NIST 800-63B), to inform platform design.

Tech stack: Python, TypeScript, FastAPI, SQLAlchemy 2.0, Alembic, PostgreSQL 16, JWT, bcrypt, React 19, Vite, Mantine, Docker Compose, AWS EC2, ECR, GitHub Actions

SRS Conversational AI Assistant

Multi-tenant RAG chatbot platform. Oversaw overall development, with a primary focus on reviewing and verifying the backend.

- Oversaw development of a production platform that gives each tenant an isolated knowledge base, a branded embeddable chat widget and a role-gated React/TypeScript admin portal (Super Admin, Tenant Admin, Support Agent), all served by a single FastAPI service.
- Reviewed and verified the backend code (~49K lines of Python across API, service and repository layers) for correctness, tenant isolation and security before release.
- Verified the document-ingestion pipeline: parsing of files, URLs and cloud sources, SHA-256 duplicate detection, structure-aware chunking, embeddings (OpenAI or local sentence-transformers) and pgvector storage, with layout-aware PDF parsing (Docling) and OCR (RapidOCR).

- Reviewed the hybrid retrieval layer, which combines pgvector similarity with BM25 via Reciprocal Rank Fusion, strictly scoped to each tenant, along with query reformulation and three-tier conversation memory.
- Verified answer-quality and safety controls: grounding checks, LLM claim verification, prompt-injection and jailbreak defence, and secret/PII output filtering, with sensitive content excluded from retrieval at ingestion.
- Checked security and multi-tenancy controls across the backend: server-enforced tenant scoping on every query, RBAC, rotating refresh tokens, account lockout, SSRF protection and GDPR retention/purge workflows.
- Oversaw delivery of cloud and SaaS source integrations (S3, Google Drive, Confluence, Jira, Slack and others), live hand-off to human agents over WebSockets, and a Finance Assistant using validated, read-only text-to-SQL.
- Verified changes against an enforced 80% pytest coverage gate (130+ test modules), with Docker builds deployed to AWS through GitHub Actions.

Tech stack: Python 3.11, TypeScript, FastAPI, Uvicorn, SQLAlchemy, Alembic, Pydantic, PostgreSQL, pgvector, OpenAI / Azure OpenAI, Docling, RapidOCR, sentence-transformers, WebSockets, APScheduler, React 19, Vite, Tailwind CSS, shadcn/ui, TanStack Query, Docker, AWS EC2, ECR, RDS, SSM, GitHub Actions, pytest

Inventzo Systems

June 2022 – May 2026

SDLC automation, Jira integration and architecture

- Engineered “Story-Assist”, a Python/FastAPI tool that streamlines the SDLC through bulk creation and linking of Jira stories to Epics and Features.
- Developed a custom version control system for user stories with state management and rollback to protect data integrity during requirement gathering.
- Automated Jira workflows with backend services for sub-task generation and bidirectional linking, reducing manual admin for the engineering team.
- Designed and deployed scalable REST APIs with FastAPI and Uvicorn, with robust error handling and debugging protocols.
- Implemented secure SSO for internal tools across the SDLC platform.
- Wrote and refined complex SQL queries to manage project metadata and user story history.
- Integrated a RAG pipeline that generates user stories, acceptance criteria and task descriptions from historical Jira data and requirement documents.
- Implemented vector-database semantic search to retrieve similar past stories, improving reuse, consistency and accuracy.
- Built and fine-tuned AI/ML models for requirement analysis, story classification and effort estimation.
- Developed prompt-engineered LLM workflows that give context-aware story suggestions.
- Managed end-to-end Git version control for the project, including complex branching strategies and merge-conflict resolution.

Tech stack: Python, NumPy, Pandas, PostgreSQL, Apache Superset, Postman, AWS, FastAPI, LLM, AI/ML, RAG, Vector DB

Python AI/ML developer, data analytics

Database Access Management (DAM): an auditing and monitoring platform that captures every SQL query, access event and user action across multiple databases

- Designed AI-driven dashboards for real-time visualisation and decision support using Dash, Plotly and Panel.
- Built and deployed ML models to analyse behavioural patterns, detect anomalies and classify user actions from historical logs and real-time query streams.
- Trained and evaluated supervised Scikit-learn models for event classification and anomaly detection.
- Engineered automated Pandas/NumPy pipelines to clean, transform and label training data.
- Developed pytest suites validating ML pipelines, data transformations and dashboard API endpoints.
- Implemented a RAG framework for natural-language querying of historical database logs and audit trails.
- Indexed SQL queries, activity logs and access patterns with embeddings for fast similarity search and threat investigation.
- Used prompt-engineered LLMs to generate insights, anomaly explanations and recommended actions for suspicious behaviour.
- Enriched ML models with RAG context, improving anomaly detection accuracy and reducing false positives.

Tech stack: NumPy, Pandas, Scikit-learn, Plotly, Dash, Panel, SQLite, SQL Server, MySQL, pytest, AI/ML, RAG

Network discovery tool and password masking application

- Developed a multi-threaded Python network discovery engine to scan, identify and monitor servers, IPs and network resources in real time.
- Implemented socket-based communication and system-level integrations to gather host and network metadata.
- Built password masking and encryption with industry-standard cryptographic libraries to protect credentials in storage and transit.
- Developed the Password Mask Lite browser extension for automated form filling and secure credential handling.
- Optimised performance with multithreading and asynchronous processing for minimal latency.

Tech stack: Python, SQL, Browser extensions, Multithreading

SIDE PROJECTS

PPCs of India

Philately reference website, started during my internship · praneethsaichunduru.github.io/ppcsofindia.github.io

- Designed, built and maintain a static website cataloguing India's Permanent Pictorial Cancellations (PPCs) and special cancellations by state and union territory, hosted on GitHub Pages.
- Grew it to 480+ pages: a page per PPC post office with the cancellation, its subject and related stamps, plus a PIN code index, special-cancellation listings for 2023–2025 and stamp maps.
- Built the shared header and footer as Web Components that render the same from a server or locally, with a light/dark theme, share links and Google Analytics.
- Compiled the site's content into a pocket book published on Amazon.

Tech stack: HTML, CSS, JavaScript, Web Components, GitHub Pages, Google Analytics

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL, PL/SQL
Backend and APIs	FastAPI, Flask, SQLAlchemy, Pydantic, Uvicorn, REST APIs, WebSockets, APScheduler, Alembic, Multiprocessing, Multithreading
Databases	PostgreSQL, pgvector, Oracle, SQL Server, MySQL, SQLite
AI, ML and NLP	RAG, Vector databases, Embeddings, OpenAI, Azure OpenAI, Prompt engineering, Hybrid search (BM25 + RRF), Text-to-SQL, Docling / OCR, Scikit-learn, NLP
Data engineering and ETL	Informatica PowerCenter, ETL pipelines, Data integration, Pandas, NumPy, PySpark
Frontend	React 19, Vite, Tailwind CSS, shadcn/ui, TanStack Query, Mantine
Security	RBAC, JWT, bcrypt, Fernet, SSO, Multi-tenant isolation, Prompt-injection defence, PII filtering, Audit logging
Integrations	Jira, Confluence, Slack, GitHub, Salesforce, HubSpot, AWS S3, Google Cloud Storage, Azure Blob, OneDrive, Google Drive, Dropbox
Cloud and DevOps	AWS EC2, ECR, RDS, SSM, Docker, Docker Compose, GitHub Actions, Git
Testing	pytest, pytest-cov, Postman
BI and visualisation	Apache Superset, Dash, Plotly, Panel, Recharts
Tools and platforms	VS Code, Visual Studio, MySQL Workbench, Microsoft Visio, Windows, Unix/Linux

RESEARCH AND PUBLICATIONS

Cross-platform web security solutions using deep learning

EDUCATION

B.Tech in Computer Science Engineering · PES Institute of Technology, Bengaluru